Probability Live Q&A (Day 1)

Quick recap

This was a live Q&A session following a probability lecture, where Hyungsuk answered student questions about statistical concepts. Breanna asked about probability interpretation with continuous distributions, and Hyungsuk explained how the sum of infinitely many probabilities with mass 0 can approach 1 as a limit. Pawan inquired about the law of large numbers and central limit theorem, with Hyungsuk clarifying that the convergence happens as the sample size approaches infinity and explaining how these theorems work with both point estimation and distribution approximation. Hrishabh raised questions about the law of total probability example involving fair and defective coins, and about variance calculations for different distributions, which Hyungsuk addressed by explaining the proper probability calculations and clarifying misconceptions about the examples.

Next steps

Hyungsuk

Follow up with Evan regarding the reference or study about the micrometer/Hibbury's collision rate example for launching space objects, and update the group if reference is found.

Answer Hrishabh's question about the law of total probability example (slide 36) after reviewing the specific slide, and provide clarification during the next session or via Slack.

Collaboration

All attendees: Post any further questions about the lectures on Slack for follow-up by Hyungsuk, PAs, or instructors.

Summary

Financial Discussion and Lecture Inquiry

Hyungsuk discussed a financial amount of $37,000 and $61,000, and mentioned being on behalf of Dr. Dave Winter. He inquired about the probability lecture and asked if participants had any questions about it. There was also a message intended for Skylar, though the content of that message was not specified in the transcript.

Probability and Statistical Concepts Discussion

The discussion focused on probability concepts and statistical theories. Breanna asked about probability as 1 over infinity, which Hyungsuk explained using the intuition of infinitely many numbers with mass where the sum approaches one. Pawan inquired about the law of large numbers and central limit theorem, with Hyungsuk clarifying that the convergence happens as n approaches infinity, which can be interpreted as approaching the actual population size in practical applications. The group also discussed that while live Q&A sessions are not recorded, daily summary PDFs are created and shared with attendees.

Law of Total Probability Example

Hrishabh asked about a confusion regarding the law of total probability example involving 5 fair coins and 1 defective coin, where he calculated a probability of 7/12 but realized his approach was incorrect. Hyungsuk explained that the law of total probability requires considering conditional probabilities based on which coin is selected, rather than simply counting total heads and tails across all coins. The discussion was cut off as they were reviewing slide 36 to examine the specific example in more detail.

Probability Concepts Discussion Session

Hyungsuk and Hrishabh discussed probability concepts, including the difference between frequentist and Bayesian approaches. They examined a problem involving coin tosses and calculated probabilities, with Hyungsuk explaining the frequentist approach in detail. Hrishabh also asked about variance calculations for two different distributions, and Hyungsuk provided a step-by-step explanation showing how the variance of 1 for the first distribution and 100 for the second distribution was calculated. The session concluded with Hyungsuk offering to answer additional questions on Slack or in a follow-up Q&A session later that day.

Statistical Inference Live Q&A (Day 1)

Quick recap

This was a live Q&A session for statistical inference led by Hyungsuk, where students asked questions about asymptotic normality, confidence intervals, maximum likelihood estimation (MLE), and variance estimation in astronomical data analysis. Pawan inquired about the connection between asymptotic normality and the central limit theorem, which Hyungsuk confirmed was the theoretical foundation for the optimal properties of MLE. Marios asked about estimating H0 and variance in the Hubble constant example, and

Hyungsuk explained that the model jointly optimizes both parameters to account for measurement error and model uncertainty. George discussed the advantages of joint optimization over conditional optimization approaches and asked about extending the model to handle different variances for measurement groups. Pawan also asked about non-probabilistic methods like fuzzy logic in astronomical data analysis, and Hyungsuk responded that while probabilistic models are preferred for uncertainty quantification, fuzzy logic could potentially be useful in clustering applications. The session concluded with discussions about bias-variance tradeoffs, model selection, and the importance of proper variance estimation to avoid introducing bias in parameter estimates.

Next steps

Hyungsuk

- Try to find the answer to the question about connecting R to NASA archive and post the answer on Slack.

- Upload the annotated lecture notes to the daily summary or the folder containing the lecture materials.

Collaboration

- All participants: Leave any further questions on Slack for Hyungsuk to monitor and answer.

Summary

Statistical Inference Q&A Session

Hyungsuk conducted a live Q&A session on statistical inference. Pawan asked about asymptotic normality, and Hyungsuk explained that it comes from the central limit theorem and shares the same conditions as the maximum likelihood estimation. Lee inquired about connecting to R3/NASA real-time data, which Hyungsuk acknowledged as a good question and requested to be posted on Slack for follow-up.

Confidence Intervals with MLE

Hyungsuk explained the construction of confidence intervals using maximum likelihood estimation and the optimal properties of MLE, emphasizing that the interpretation depends on the construction method rather than specific data calculations. He distinguished this approach from Bayesian inference by highlighting that the confidence interval's nominal level is guaranteed through the asymptotic distribution of the maximum likelihood estimator in a hypothetical repeated sampling scenario.

h-null and Measurement Variance Estimation

Hyungsuk discussed the estimation of h-null and the variance of measurement using Hoppe data, explaining that the model contains two unknown parameters: h-null and sigma squared, which represents the variance of the measurement error. He distinguished between statistical uncertainty (quantified by sigma square) and model uncertainty (quantified by the inverse Hessian matrix), noting that they are jointly optimized rather than optimized conditionally. Hyungsuk confirmed that the notation used in the discussion was correct.

Maximum Likelihood vs Least Squares

George asked about the difference between maximum likelihood estimation and traditional least squares methods, specifically regarding the inclusion of variance estimates. Hyungsuk explained that maximum likelihood allows for joint optimization of parameters, which is superior to conditional optimization approaches that estimate variance separately, though he noted that in practice some researchers use iterative processes with marginal distributions. The discussion concluded with Hyungsuk confirming he had addressed Mario's previous questions and preparing to continue with the lecture.

Variance Modeling in Astronomy

George expressed appreciation for Hyungsuk's approach to modeling variance, noting that the optimal value of H0 appears to work across a range of possible variance values, which he found impressive. Hyungsuk explained that the model includes uncertainty in measured quantities like velocity and distance to avoid overconfidence in Hubble constant estimates, and mentioned that astronomers commonly add extra uncertainty beyond reported one-sigma error bars.

Measurement Error Modeling Discussion

Hyungsuk and George discussed modeling measurement errors and variance estimates, with George suggesting the possibility of using separate variances for different groups of measurements based on different conditions. Hyungsuk explained that this approach could be more sophisticated than the simple model currently used and could account for heteroscedastic measurement errors across groups. They also briefly touched on how Hubble might have handled similar calculations in his 1929 paper, though the details remain unclear. The conversation concluded with a mention of Tom's upcoming Bayesian inference class on Wednesday.

Bias-Variance Tradeoff Discussion

Pawan asked about the bias-variance tradeoff and whether reducing variance might increase bias, which Hyungsuk explained involves finding a balance between model complexity and prediction ability using cross-validation techniques. Hyungsuk clarified that while more complex models reduce bias, they can increase variance and prediction errors, requiring trial and error to find the optimal balance. Pawan also inquired about non-probabilistic models like fuzzy logic for astronomical datasets, though this topic was not directly addressed in the discussion.

Astronomical Data Modeling Techniques

Hyungsuk and Pawan discussed using probability language and fuzzy logic for modeling astronomical data, particularly for uncertainty quantification and clustering applications. They explored how fuzzy logic could handle measurement noise and outliers in astronomical data, with Pawan suggesting applications in clustering and unsupervised learning. Hyungsuk explained the mathematical rationale for using maximum likelihood estimation despite uncertainty about true parameter values, noting its asymptotic properties make it suitable for practical data analysis.

Handling Bias and Variance in Data

The group discussed methods for handling bias and variance in data analysis, with Hyungsuk and George agreeing that combining physical models with uncertainty and statistical models would be ideal for addressing scientific questions. They explored the use of well-defined estimators like Bayesian posteriorism and maximum likelihood estimation to better understand bias properties. The discussion also touched on robust statistics and model misspecification, with Hyungsuk mentioning recent literature on machine learning techniques for handling these issues. Andrew indicated he would close the meeting, and Hyungsuk encouraged participants to leave any remaining questions on the slide for future discussion.